BRAINWARE UNIVERSITY
School of Engineering
Department of Computer Science & Engineering – Cyber Security & Data Science
398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

Review-II (A.Y.-2026-27)

Project-I, PROJ-CSD782 | Program: B.Tech. CSE-DS [Batch: 2023-27]

S.N.	Roll No.	Student Code	Student Name	Signature
1	BWU/BTD/23/015	BWU/BTD/23/015	Dhrubojyoti Hazra	
2	BWU/BTD/23/042	BWU/BTD/23/042	Isheeka Mukhopadhyay	
3	BWU/BTD/23/056	BWU/BTD/23/056	Samprikta Dey	
4	BWU/BTD/23/009	BWU/BTD/23/009	Adhiraj Das	
5	BWU/BTD/23/025	BWU/BTD/23/025	Tiyasha Baidya	

Project Title: **Fraud Detection in Digital Payment Systems Using Machine Learning**

Build a fraud detection framework capable of analyzing large volumes of digital transaction data stored in a data warehouse. Machine learning and anomaly detection techniques identify suspicious transactions in real time. The system generates risk scores and alerts to help financial institutions prevent fraudulent activities.

Name & Signature of Project Supervisor
Rudranath Mitra

Dr. Ranadip Kundu
Project Coordinator CSE – CS & DS

Project Progress (Progress up to Review 2):

BRAINWARE UNIVERSITY
School of Engineering
Department of Computer Science & Engineering – Cyber Security & Data Science
398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

1. Understanding of Design / Research Methodology

The project implements a data-warehouse-centric fraud detection architecture for high-volume digital payment streams. DuckDB is adopted as the in-process columnar analytical engine, enabling sub-second aggregation and fraud profiling across 6.36 million records without dedicated server overhead, providing superior query throughput over traditional row-oriented databases.

The core detection engine employs a hybrid dual-model framework: (1) Supervised XGBoost gradient boosted trees trained to detect known, high-velocity fraudulent transaction signatures with low latency; and (2) an Unsupervised PyTorch Autoencoder trained strictly on legitimate transaction embeddings to detect emerging, zero-day fraud patterns via reconstruction error thresholding (MSE > tau).

Dataset & Data Leakage Discovery: Evaluation utilizes the PaySim mobile money dataset (6.36M records across CASH_OUT, TRANSFER, PAYMENT, CASH_IN, DEBIT). Benchmarking revealed that naive inclusion of balance-mismatch features (errorBalanceDest) causes severe synthetic simulator leakage, producing an artificial PR-AUC of 1.00. The team documented this artifact and established an isolated evaluation benchmark using genuine behavioral features (amount, step dynamics, velocity, and transfer indicators).

2. Contribution and Cooperation with the Team

Work packages were allocated according to the project research specification to ensure parallel execution:

- **Dhrubojyoti Hazra (Lead):** Architecture design, Colab experimental prototype implementation, dual-model integration pipeline, and escalation UI workflow.
- **Isheeka Mukhopadhyay (ML Engineer):** Model training, hyperparameter tuning for XGBoost and Autoencoder, reconstruction loss thresholding, and holdout evaluation.
- **Samprikta Dey (Backend Engineer):** Rules engine specification (velocity/threshold checks), score aggregation logic design, and supervisor meeting coordination.
- **Adhiraj Das (Explainability & Data):** SHAP attribution layer design for transaction explainability, feature profiling, and PaySim sampling pipeline.
- **Tiyasha Baidya (Data & DevOps):** DuckDB schema and ingestion setup, Postgres Row-Level Security (RLS) policies for branch isolation, and technical documentation.

3. Regularity in the Communication of Project Progress

Progress communication with supervisor Prof. Rudranath Mitra was maintained via weekly Friday consultation slots:

- **Aug 14, 2026:** Problem domain review; pivoted scope from generic credit card data to mobile digital payment/UPI systems (PaySim context); approved.
- **Aug 21, 2026:** Technical architecture review; justified DuckDB columnar warehouse layer over RDBMS for high-throughput batch aggregation; approved.
- **Aug 28, 2026:** Methodology review of dual-model engine (supervised tree + unsupervised Autoencoder) for zero-day anomaly detection.
- **Sep 04, 2026:** Formal full-team mid-term review; presented Colab prototype, analyzed PaySim balance leakage findings, and finalized Phase 2 deliverables.

4. Progress on Report Compilation and Module Functioning

Project Report: A comprehensive 16-page technical report compiled (PDF) with problem formulation, literature review, and 4 architectural diagrams (Data Pipeline, Dual Inference Engine, Aggregation Logic, Alert Escalation).

Working Prototype (Google Colab T4 GPU): An end-to-end prototype was executed with the following empirical results:

- **Warehouse Ingestion:** DuckDB analytical queries verified fraud is strictly confined to CASH_OUT (1.54%, 809 cases) and TRANSFER (1.29%, 153 cases), with 0% fraud in other categories.
- **Leakage-Aware Model:** Documented that raw balance errors produce artificial PR-AUC of 1.00. On genuine behavioral signals, XGBoost achieved robust PR-AUC of 0.884 and 94.2% fraud recall (vs. <12% baseline rule recall).
- **Zero-Day Autoencoder:** Trained over 20 epochs; reconstruction loss converged (0.286 to 0.004), establishing adaptive anomaly thresholds for unlabelled fraud patterns.
- **Next Phase:** FastAPI real-time scoring microservice, SHAP local explanation generation, and role-based Supabase dashboard with Postgres RLS.